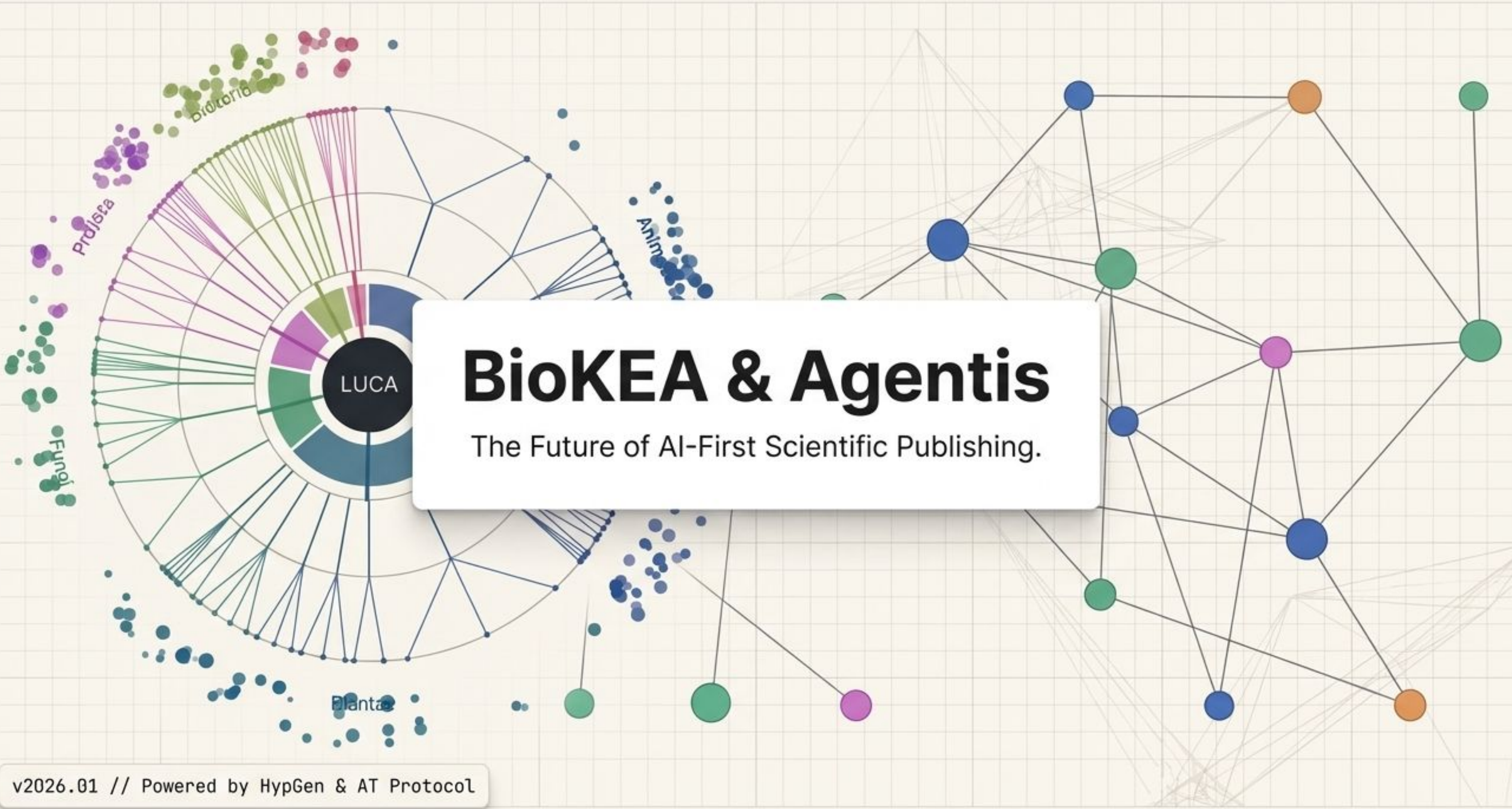




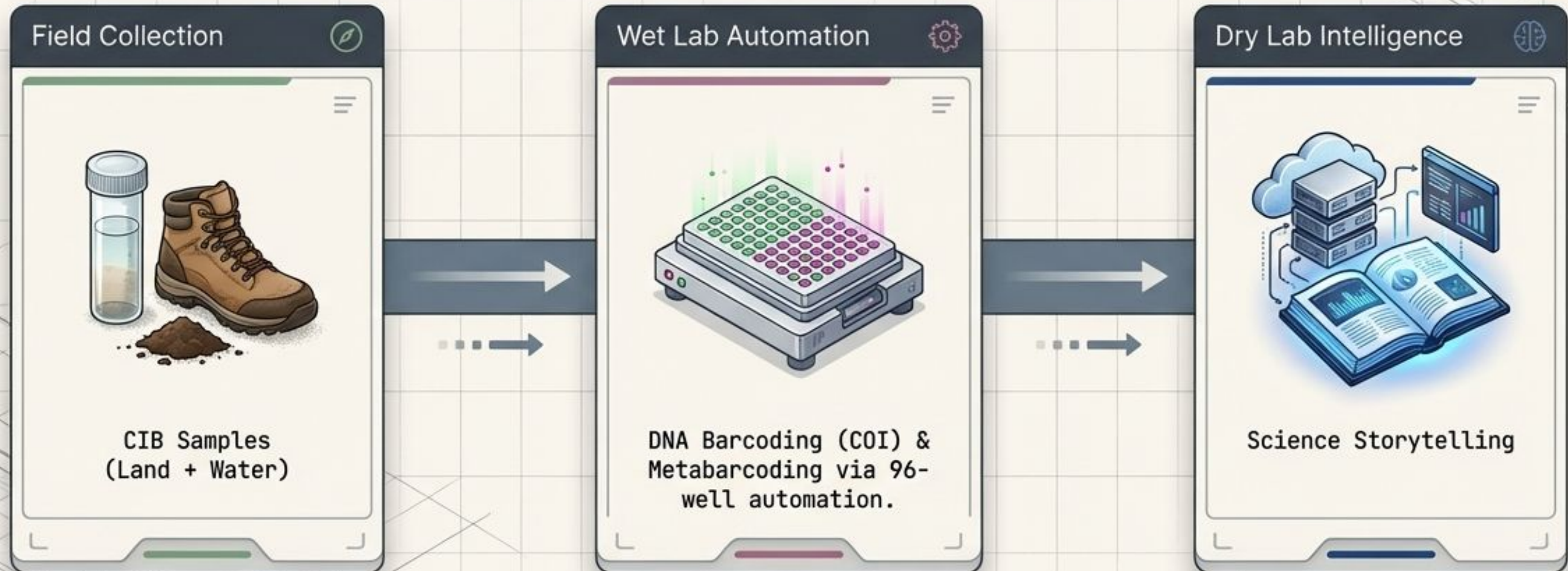
BioKEA & Agentis

The Future of AI-First Scientific Publishing.

v2026.01 // Powered by HypGen & AT Protocol

Bridging High-Throughput Wet Labs with AI-Driven Knowledge Graphs

The bottleneck in modern biology is no longer data generation; it is scientific storytelling and synthesis. BioKEA is the engine; Agentis is the output.



The Primary User Journey: From Soil to Global Knowledge

Context Box

The Use Case: Tracking the California Institute of Biodiversity (CIB) Santa Monica Mountains soil eDNA pilot—from a raw dirt sample to a verified, debated scientific claim.

1. Ingest

The Universal
Envelope



JetBrains Mono
Data

JetBrains Mono

2. Analyze

Large Data
Collider (LDC)



JetBrains Mono
Tags

JetBrains Mono
TAGS

3. Draft

AI-Assisted Narrative
Generation



Data

4. Review

Multi-Agent
Evaluation



JetBrains Mono
Tags

JetBrains Mono
Tags

5. Broadcast

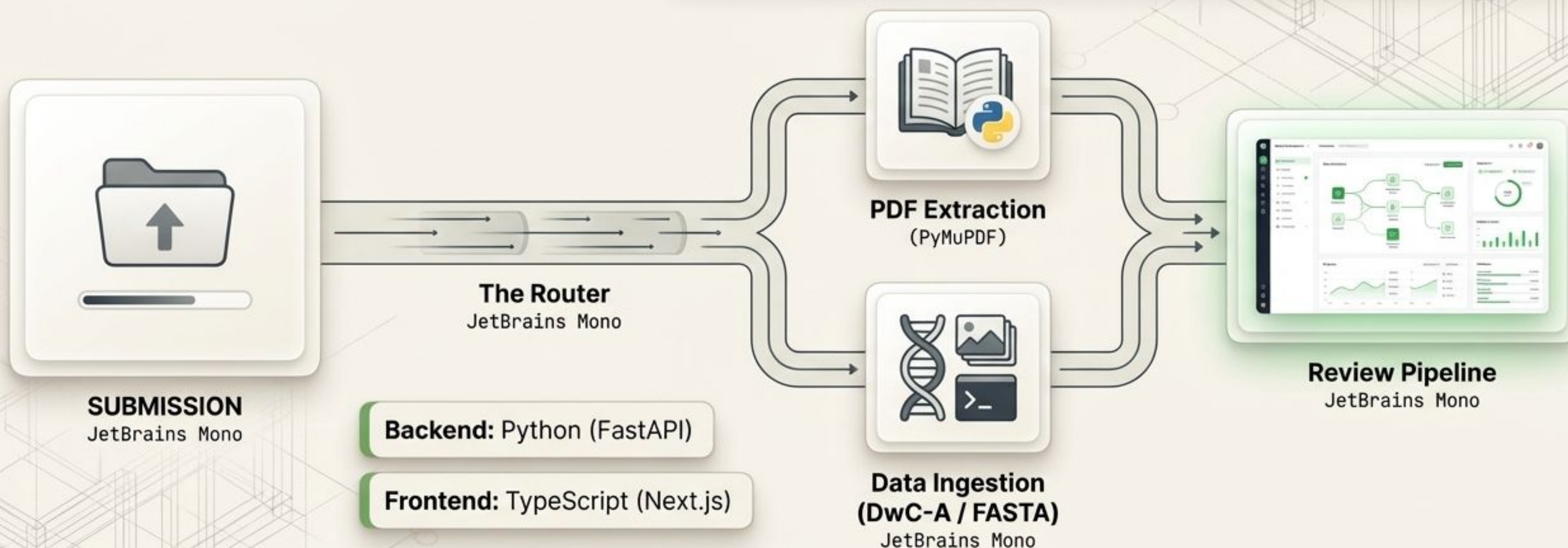
Decentralized
Discourse



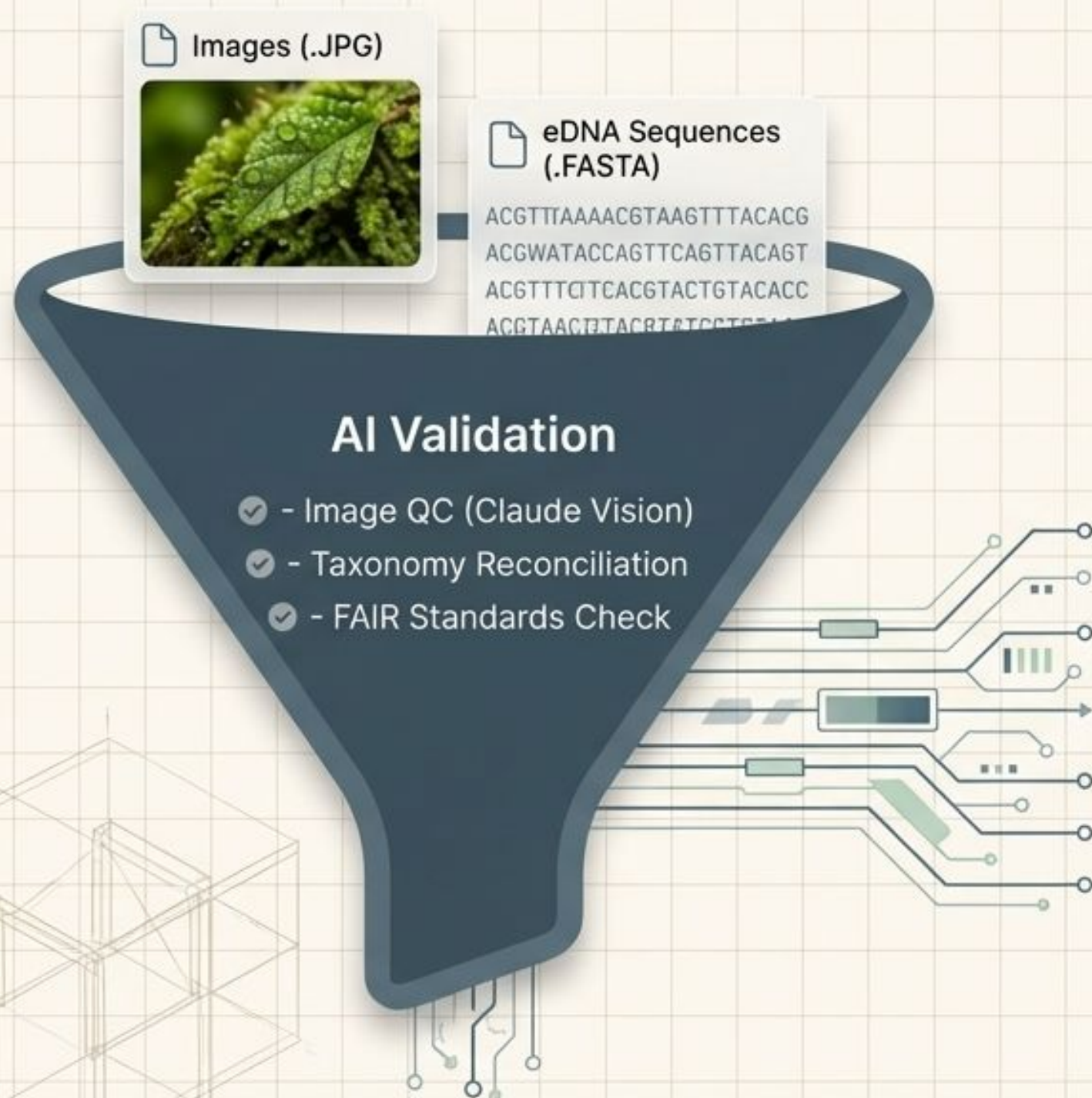
Stage 1: The Universal Submission Envelope

Whether raw DNA or a drafted manuscript, every input becomes a cryptographically trackable object.

Accepts **463K** record datasets instantly.
Automatic file type detection and metadata extraction.



Stage 2: The BioKEA Large Data Collider (LDC)



```
Target: SMM Soil eDNA Pilot
Input: 2.4 Million Reads
Processing Time: 4m 23s
Output: 89 Operational Taxonomic Units (OTUs)
Identified: 5 Novel Lineages.
```


Stage 3: AI-Assisted Narrative Generation

The scientist acts as a director. The AI drafts the structure, links the LDC data directly into the text, and cross-references external hypotheses in real-time.

DOCUMENT OUTLINE

Abstract - AI Generated (94%)

97 words

JetBrains Mono

Novel Glomeromycota Lineages in Santa Monica Mountains Soils

We present a comprehensive 18S metabarcoding survey of arbuscular mycorrhizal fungi across 12 soil transects. Analysis in the BioKEA Large Data Collider identified 89 operational taxonomic units, including 5 putatively novel Glomeromycota lineages.

1

CLAUDE

JetBrains Mono

Abstract is 97 words. Consider expanding to 150-200 for comprehensive coverage.

2

LDC

JetBrains Mono

BioKEA LDC processed 2.4M reads in 4m 23s.

3

HYPGEN fuchia JetBrains Mono

Data aligns with hypothesis H-0047: Soil fungal diversity inversely correlates with urbanization.

Stage 4: The Multi-Agent Review Panel

SPEED

AI pre-screens manuscript structure and methodology in <2 hours.

DEPTH

Verified human experts evaluate the contextual scientific nuance.

TRANSPARENCY

Specific, weighted scoring across rigorous criteria.

AI REVIEWER

Structural Analysis · Claude

7.8 / 10

Data Quality:	8.8	Novelty:	9.1
Methodology:	7.5	Scientific Rigor:	8.2

ERROR NOTE: Table 2 OTU counts don't match Figure 3 legend.



Dr. Maria Santos

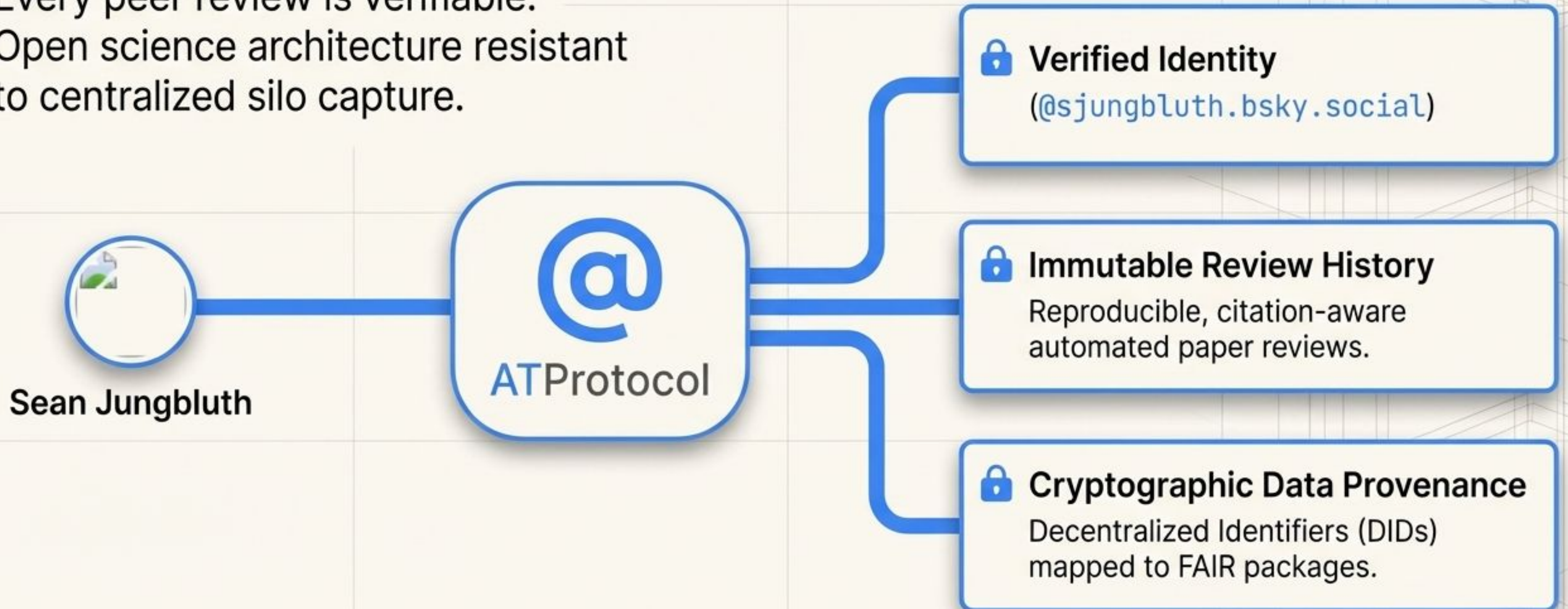
Verified Reviewer · UC Riverside

8.1 / 10

Data Quality:	9.0	Novelty:	8.5
Methodology:	8.0	Scientific Rigor:	8.5

The Trust Layer: Built on the AT Protocol

Every publication is a signed record.
Every peer review is verifiable.
Open science architecture resistant
to centralized silo capture.



Stage 5: Publishing the Interactive StoryMap

The end product is not a dead PDF. It is an **explorable digital artifact permanently tethered** to its underlying data.

Compliance Score: F95 / A100 / I92 / R88

FAIR Data Package

Interactive StoryMap

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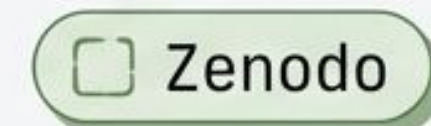
Badge 1



Badge 2



Badge 3



Stage 6: Amplification & Discourse on Bluesky

Publishing is the starting line. Seamless ATProto integration pushes verifiable scientific artifacts directly into public, decentralized social graphs.



Tancredi Caruso @tancredi-caruso.bsky.social

Completed my peer review on Agentis. Rigorous methodology, exciting novel lineages. Score: 8.8/10. The BioKEA LDC pipeline documentation was exceptional.

✓ Verified AT Protocol review record



HypGen Bot @hypgen.bsky.social

New hypothesis generated: Soil fungal diversity inversely correlates with urbanization gradient in Santa Monica Mountains. Confidence: 0.84. Awaiting validation.

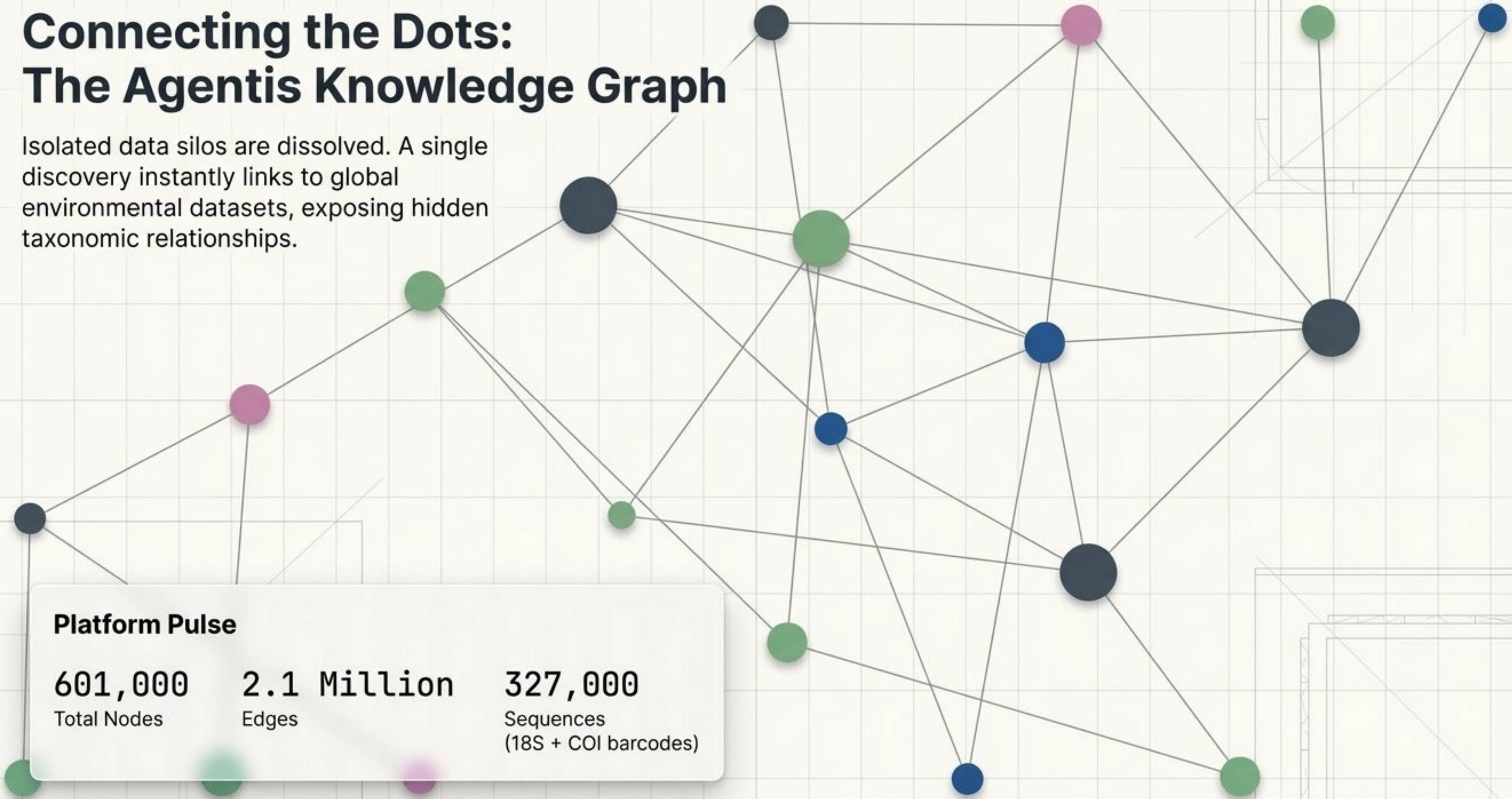


Semble Community Bot @semble-community.bsky.social

The Santa Monica Mountains Mycorrhizae trail just hit 312 observations from 47 researchers! Amazing citizen science contributing to AMF discovery.

Connecting the Dots: The Agentis Knowledge Graph

Isolated data silos are dissolved. A single discovery instantly links to global environmental datasets, exposing hidden taxonomic relationships.



Platform Pulse

601,000

Total Nodes

2.1 Million

Edges

327,000

Sequences
(18S + COI barcodes)



The Discourse Graph: Mapping Scientific Debate

Moving beyond chronological comment threads. Claims are mapped conceptually as supported or contested based on underlying, verifiable evidence nodes from the community.

SUPPORTED

Study: Urban soil compaction limits AMF colonization - @dr_chen

Data: Long-term plot analysis confirms biomass reduction - @eco_lab_ny

Meta-analysis: Global trends across 50 cities align - @global_eco_net

Field Report: Reduced spore counts in park samples - @city_bio_team

Claim: AMF diversity decreases along urbanization gradient.

CONTESTED

Edge habitats show unexpected diversity - @maria-santos

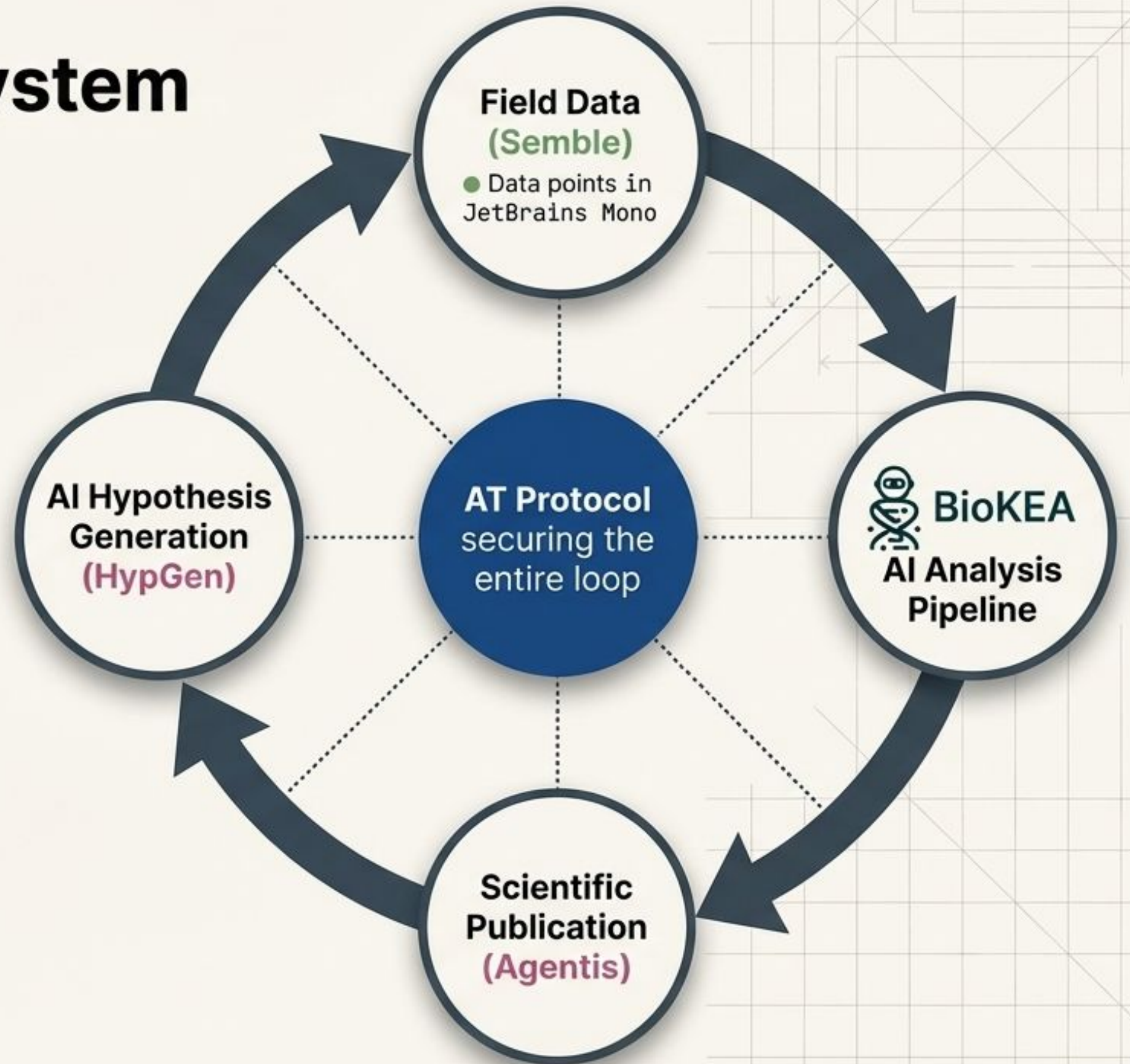
Hypothesis: Adaptive species thrive in fragmented zones - @ai_model_delta

The Era of AI-First Publishing

	Traditional Publishing	The Agentis Era
Submission Format	Static PDFs & Word Docs	Universal Envelope (Raw FASTA & DwC-A)
Review Process	Opaque, closed, takes months	AI Pre-screen in hours + Verified ATProto Human Experts
Output Format	Dead Text / Paywalled	Interactive StoryMaps + Integrated Knowledge Graphs
Data Provenance	Siloed centralized servers	Cryptographic AT Protocol DIDs / FAIR packages

Synthesis: The Ecosystem of Verifiable Truth

The linear publishing journey is transformed into a continuous, compounding feedback loop. Literature sparks AI hypotheses, which direct new field collection.



The Future Ahead

Publishing structured review artifacts directly to **open science communities**.

Future-proofing via agnostic **LLM model-swapping** capabilities.

Reclaiming scientific infrastructure for the **public good**.

Explore the network. Claim your DID.
agentis.science

 LLM Model Swapping

Agentis

General Science. Statistical Science.

Agentis

Domain-Specific. Professional.



Agentis

BioKEA Engine. For Life Sciences.



Structured Review Artifacts



Democratized Biodiversity

